

Stackly Energy – Technical Project Documentation

1. Project Title

Stackly Energy – Renewable Energy Grid Telemetry Platform Website

2. Project Description

Stackly Energy is a premium, responsive renewable energy grid monitoring and telemetry platform website developed using HTML, CSS, and JavaScript. The website provides information about smart grid integration, utility-scale solar and wind services, local savings calculations, carbon offsets, and specialized dashboards for consumers and platform administrators.

3. Technologies Used

Frontend

- **HTML5:** Semantic document structuring and inline SVG vector diagrams.
- **CSS3:** Variable-driven layout styling, neon color tokens, custom scrollbars, keyframe animations, and glassmorphic panels.
- **JavaScript (ES6):** Real-time form validations, search filter algorithms, interactive hamburger toggles, and live telemetry rendering loop timers.

Libraries & Design Assets

- **Font Awesome Icons:** Lightweight iconography for navigation links.
- **Google Fonts (Outfit):** Premium, modern sans-serif typography.

Deployment & Versioning

- **GitHub:** Repository management, tracking history, and branching.
 - **GitHub Pages:** Static hosting of the production environment.
 - **Browser Cache Invalidation:** Dynamic query parameter versioning (`?v=2.4`).
-

4. Project Modules

Public Website (Marketing)

- **Home Page:** Main hero landing area featuring neon glow backdrops, key benefit grids, and animated loading fade-slides.
- **About Page:** Mission history, environmental offset metrics, and key milestones timeline.
- **Services Page:** Sizing calculator (local utility savings estimator) and clean energy service catalogs.
- **Blog Page:** Technical publications detailing predictive ML grids, line loss audits, and PV element recycling.
- **Contact Page:** Interactive contact inquiry form, office location coordinates, and support routing.
- **Support Pages:** FAQs, Team roster, Testimonials, Statistics, and a custom **404 page**.

Authentication

- **Login Page:** Secure login gateway to access consumer/administrator control panels.
- **Register Page:** User registration form to sign up for telemetry monitoring access.

User Dashboard (Consumer)

- **Dashboard Overview:** Telemetry node registries, live battery storage statistics, and real-time load consumption.
- **My Projects:** Installed panel counts, system capacity indicators, and active node grids.
- **Energy Reports:** Environmental offset logs, carbon credits tracking, and historic generation grids.
- **Profile Settings:** First and Last Name adjustments, email, and phone number modifications.
- **Help Support:** Support ticket submissions, helpdesk records, and status lists.

Admin Dashboard (Operator)

- **Dashboard Overview:** Overall grid capacity, active node counters, and active system operations event logs.
 - **User Management:** Consumer accounts list, carbon credit approvals, and user growth trackers.
 - **Services Management:** Service catalog additions, publishing controls, and popularity demand meters.
 - **Projects Management:** Grid station latencies, substation logs, and operational capacity meters.
 - **Settings Management:** Tariff base adjustments, database cloud backup frequencies, API tokens, and sync rates.
-

5. Module Descriptions

1. Home Module

Serves as the entry point of the Stackly Energy platform. It introduces clean energy solutions, presents high-contrast interactive call-to-actions, lists platform capabilities, and guides user flow to secondary pages.

- **Features:** Dynamic split-grid Hero section, stats counters, clean layout, responsive menu drawer.

2. About Module

Presents the technical lineage, mission statement, environmental progress metrics, and key organizational history milestones.

- **Features:** "GRID/SOL/WND/BAT" satellite SVG layout, mission features list, environmental statistics offsets.

3. Services Module

Highlights localized domestic and industrial solutions. Contains a smart calculator that processes regional tariffs and utility averages.

- **Features:** Sizing savings calculator (interactive JS logic), customized consulting anchors.

4. Blog Module

Hosts technical news, engineering logs, and research articles concerning renewable grid development.

- **Features:** 6 comprehensive blog grids, filter navigation bar, read-time calculators.

5. Contact Module

Provides communication channels to the consulting team. Real-time inputs restrict characters to guarantee clean contact details.

- **Features:** Inquiry form, contact info details, location coordinates map.

6. Authentication Module

Safeguards access controls, validates entries, and routes users to their matching administrative dashboard.

- **Features:** Role-based access redirect, password visibility controls, registration validations.

6. Features

User Features

- **User Registration & Login:** Role-based authentication routes.
- **Telemetry Node Tracking:** Live indicators of battery levels and active load consumption.
- **Energy Generation Metrics:** Daily generation curves and carbon displacement tracking.
- **Support Desk Access:** Instant ticket submissions and real-time status trackers.

Admin Features

- **Substation Monitoring:** Live database counters of operational grid load.
- **Service Sizing Publishing:** Form to release new services directly to user grids.
- **Global Tariff Adjustment:** Controls to modify base electricity rates and battery alert limits.
- **Access & Sync Frequencies:** Configuration panel for cloud backups and API access tokens.

Website Features

- **Frosted Glassmorphism:** Semi-transparent dark panels with frosted blur shadows.
- **Scroll Lock Navigation:** Freezes underlying background scrolling when mobile menu drawer is open.
- **Real-Time Input Sanitizers:** Blocks letters in numeric inputs and numbers in alphabetical inputs.
- **Proportional Logo Scaling:** Auto-scales the Stackly logo branding on all viewport resolutions.

7. Folder Structure

```
Stackly-Solar-Energy/
├─ index.html
├─ about.html
├─ services.html
├─ blog.html
├─ contact.html
├─ login.html
├─ register.html
├─ 404.html
├─ admin-dashboard.html
├─ admin-messages.html
├─ admin-projects.html
├─ admin-services.html
├─ admin-settings.html
├─ admin-users.html
├─ user-dashboard.html
├─ user-profile.html
├─ user-projects.html
├─ user-reports.html
├─ user-support.html
├─ css/
│   └─ main.css
│   └─ pages.css
│   └─ dashboards.css
├─ js/
│   └─ main.js
│   └─ pages.js
│   └─ dashboards.js
└─ images/
    └─ logo.png
    └─ [assets].webp
```

8. Security

- **Client-Side Sanitizers:** Real-time input filters strip illegal characters (digits/letters) automatically on key inputs.
- **Stacking Index Isolation:** Fixed overlay layering (z-index: 999 and 1000) prevents overlay clicks from bypassing active control menus.

- **Session-Based Navigation Routing:** Segregates admin controls from standard consumer paths.
-

9. Responsive Design

The platform supports standard device resolutions:

- **Desktop Devices:** Full grid grids, wide dual-columns.
 - **Tablets & Mobile Devices:** Single-column vertical stacking, left/right collapsible drawers.
 - **Responsive Breakpoints:**
 - 1200px (Desktop grid boundary)
 - 991px (Mobile menu activation threshold)
 - 768px (Tablet vertical columns collapse)
 - 576px (Mobile input forms vertical collapse)
-

10. Future Enhancements

- **Real-Time Solar Weather API:** Integrates weather forecasting algorithms to predict grid output deflection.
 - **Payment Gateway Integration:** Net billing tariff invoice payment channels for active consumers.
 - **Database Authentication Backend:** Dynamic user records handling using server-side databases (NodeJS/Express/PostgreSQL).
 - **Automated Carbon Certificate Generation:** Programmatically issues downloadable offset credentials to consumers.
-

11. Conclusion

Stackly Energy represents a state-of-the-art, high-aesthetic clean energy administration platform. Built on responsive UI/UX principles, glassmorphism design styles, interactive validation structures, and scroll locking, it demonstrates robust frontend execution tailored to renewable energy tracking and services catalog management.